

Yuanman Mu

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EDUCATION

Boston University, College of Arts and Sciences | Boston, MA

Sept 2021 – May 2025

Bachelor of Arts in Computer Science, Minor in Applied Statistics | GPA: 3.56/4.0

- **Honor:** Dean's List (Fall 2023, Spring 2023, Fall 2022, Spring 2022)
- **Core Courses:** Software Engineering, MEAN Stack Application Development, Mobile App Development, Computing Systems, Database Systems, Theory of Computation, Analysis of Algorithms, Data Science in R

PROJECTS

Full-stack Android App Development | Team Leader

Dec 2024

- Designed a responsibly designed full-stack Android App for English vocabulary learning using latest Jetpack Compose
- Used Moshi and Retrofit with Coroutines to handle API integrations, and used Firebase for authentication
- Implemented Room with cloud synchronization for real-time updates of studying progress across device
- Built UI with Material 3, multi-screen flows, LiveData, and ViewModel
- App tested, including Unit test, UI test, and Error Logging

Impact Analysis: Mayor's Office of Returning Citizens, City of Boston | Team Leader

May 2024

- Developed a comprehensive web dashboard analyzing the impact of the Office of Returning Citizens (ORC) in FY22-24
- Led weekly meetings, client meetings, data analysis, dashboard backend development, database schema
- Converted textual data into interactive numeric analysis using Python and NLP, producing charts and graphs for analysis
- Analyzed the statistical impact from the performance of ORC, and generated interactive graphs for ORC's annual report to the Mayor's Office

Racial and Economic Disparities Analysis in Food Accessibility | Team Leader

Mar 2024

- Analyzed food accessibility disparities across 72,532 U.S. census tracts using USDA's Food Access Research Atlas, reducing 147 variables by 60% through data cleaning
- Explored the relationship between racial demographics, poverty rates, and supermarket accessibility, identifying a notable 0.53 correlation between African American populations and poverty
- Integrated state GDP data to contextualize findings, improving model precision (R^2 increased from 0.19 to 0.24)
- Visualized key patterns via scatter plots, heatmaps, and regression analysis, highlighting systemic inequalities in food access and informing potential policy interventions
- Critically evaluated model limitations and proposed future work incorporating longitudinal and qualitative data

Spotify Library-Based Web Application | Team Leader

Dec 2023

- Collaborated with a team of 4 and developed a fully functional web application with a dynamic JavaScript-based front-end, an Express.js based back-end, a MongoDB database that stores back-end application state
- Coded most of the backend & API integration & database interactions

Comparative Rent Price Analysis Across BU Neighborhoods | Team Leader

Dec 2022

- Organized team members for different jobs such as hypotheses establishing, data collection, data analysis
- Analyzed cross-sectional data with variables room size, year built, bed types, days on markets, crime rate, and average income to understand their influences on rent price
- Found significant positive relationships between year built, bed types, room size and rent price

PROFESSIONAL EXPERIENCE

Client Services & Support | Boston University IT Help Center IS&T | Boston, MA

Sept 2023 - May 2025

- Provided technical support to students and staff, handled the delivery of devices around the campus, helped with troubleshooting hardware and software issues during deliveries
- Developed problem-solving skills and improved communication of technical information

Academic Mentor | Boston University Bridge to BU Program | Boston, MA

Sept 2021 - Dec 2022

- Supported international students' adjustment to BU through English practice, cultural discussions, and team activities
- Enhanced communication and leadership skills by guiding mentees in academic development

RESEARCH EXPERIENCE

Analysis of Machine Learning Techs | CIEI | Nanjing, China

July 2023

- Conducted cooperative research analyzing the performances of CNN, ANN, GRU, LSTM, ARIMA, and Transformer models in forecasting stock prices
- Contributed to findings that indicated LSTM outperformed other algorithms
- Produced a paper "Comparative Analysis of Different Machine Learning Techniques in Forecasting Stock Price"

TECHNICAL SKILLS

Programming Languages: Python (Excellent), Java (Excellent), JavaScript (Excellent), HTML (Excellent), R (Excellent), SQL (Good), Kotlin (Good), C (Good)

Familiar with: Next.js, React, Node.js, Jetpack Compose, noSQL, MongoDB, DBMS, XML